

Julien Siems

 [Website](#)

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EDUCATION

University of Freiburg

Freiburg, Germany

PhD Student – Machine Learning Group, supervised by Prof. Frank Hutter

Nov 2023 – Present

- Research focus: in-context learning and state-tracking abilities of linear RNNs such as Mamba and DeltaNet.
- Organizer of the In-Context Learning Workshop at ICML 2024.
- Thesis submitted July 2026.

University of Freiburg

Freiburg, Germany

Master's in Computer Science; GPA: 4.0 (US scale); Graduated with Distinction

Oct 2018 – Mar 2021

Courses: Numerical Optimization, Optimal Control, Automated Machine Learning, Game Theory

University College London

London, United Kingdom

Erasmus Exchange Student

Sep 2016 – Jun 2017

Courses: Probabilistic Graphical Models, Image Processing, Information Retrieval, Computer Vision

TU Dresden

Dresden, Germany

Bachelor's in Computer Science; GPA: 3.5 (US scale)

Oct 2014 – Dec 2017

EXPERIENCE

Prior Labs

Berlin, Germany

AI Scientist

Jan 2026 – Aug 2026

Contributed to the architecture, synthetic data prior, and time-series capabilities of the new TabPFN v3 model.

Microsoft Research

Cambridge, UK

Research Intern

Sep 2025 – Nov 2025

Authored the ICLR 2026 workshop paper “Learning State-Tracking from Code Using Linear RNNs,” whose findings, alongside “Unlocking State-Tracking in Linear RNNs through Negative Eigenvalues,” were adopted in the new Olmo Hybrid.

Merantix Momentum

Berlin, Germany

Machine Learning Researcher

Jun 2022 – Aug 2023

- NeurIPS 2023 conference paper on mitigating interpretability issues caused by concavity in generalized additive models: This project focused on addressing concavity, the nonlinear equivalent of multicollinearity, which arises in generalized additive models, specifically Neural Additive Models and NeuralProphet.

TU Munich

Munich, Germany

Research Assistant – Machine Learning and Inverse Problems Group

Nov 2021 – Apr 2022

- Adversarial training for inverse problems: My project focused on the optimization of inverse problems under provably maximally adversarial attacks. My contributions included mathematical and empirical analysis of the problem for a linear model.

University of Zurich

Zurich, Switzerland (Remote from Berlin)

Research Assistant – Computation and Economics Research Group

Apr 2021 – Sep 2021

- Iterative Combinatorial Auctions: I co-authored several publications on machine-learning-powered iterative combinatorial auctions, which are used to allocate mobile-frequency spectrum such as for 5G and 6G.

Amazon Web Services

Berlin, Germany

Applied Science Intern

Sep 2020 – Mar 2021

- Dynamic Neural Network Pruning: My internship in the AWS SageMaker team focused on AutoML and dynamic network pruning. Our proposed neural network weight-pruning method makes neural networks more efficient to evaluate, reducing energy costs.

PUBLICATIONS (* DENOTES EQUAL CONTRIBUTION)

- Julien Siems*, Riccardo Grazi*, Korbinian Pöppel, Jaisidh Sing, Arber Zela, Timur Carstensen, Jenia Jitsev, Frank Hutter, Volkan Cevher, Antonio Orvieto, Aaron Klein. “Complex KDA: Understanding and Enhancing the Expressivity of Kimi-Delta-Attention”, Under review, 2026
- Léo Grinsztajn et al. “TabPFN-3: Technical Report”, 2026
- Julien Siems*, Riccardo Grazi*, Kirill Kalinin, Hitesh Ballani, Babak Rahmani*. “Learning State-Tracking from Code Using Linear RNNs.” *LIT Workshop at ICLR 2026*.
- Vladyslav Moroshan*, Julien Siems*, Arber Zela, Timur Carstensen, Frank Hutter. “TempoPFN: Synthetic Pre-training of Linear RNNs for Zero-shot Time Series Forecasting.” *AI for Tabular Data Workshop at EurIPS 2025*.
- Julien Siems*, Timur Carstensen*, Arber Zela, Frank Hutter, Massimiliano Pontil, Riccardo Grazi*. “DeltaProduct: Improving State-Tracking in Linear RNNs via Householder Products.” *NeurIPS 2025*.
- Riccardo Grazi*, Julien Siems*, Jörg Franke, Arber Zela, Frank Hutter, Massimiliano Pontil. “Unlocking State-Tracking in Linear RNNs through Negative Eigenvalues.” *ICLR 2025 (Oral)*.
- Sathya Kamesh Bhethanabhotla*, Omar Swelam*, Julien Siems, David Salinas, Frank Hutter. “Mamba4Cast: Efficient Zero-Shot Time Series Forecasting with State Space Models.” *Workshop on Time Series in the Age of Large Models at NeurIPS 2024*.
- Andreas Müller*, Julien Siems*, Harsha Nori, David Salinas, Arber Zela, Rich Caruana, Frank Hutter. “GAMformer: In-Context Learning for Generalized Additive Models.” *Interpretable AI Workshop & Tabular Representation Learning Workshop at NeurIPS 2024*.
- Riccardo Grazi*, Julien Siems*, Simon Schrod, Thomas Brox, Frank Hutter. “Is Mamba Capable of In-Context Learning?” *ME-FoMo Workshop at ICLR 2024 & AutoML Conference 2024*.
- Julien Siems*, Konstantin Ditschuneit*, Winfried Ripken*, Alma Lindborg*, Maximilian Schambach, Johannes S. Otterbach, Martin Genzel. “Curve Your Enthusiasm: Concurrency Regularization in Differentiable Generalized Additive Models.” *NeurIPS 2023*.
- Julien Siems, Maximilian Schambach, Sebastian Schulze, Johannes S. Otterbach. “Interpretable Reinforcement Learning via Neural Additive Models for Inventory Management.” *AI4ABM Workshop at ICLR 2023*.
- Jakob Weissteiner*, Jakob Heiss*, Julien Siems*, Sven Seuken. “Bayesian Optimization-Based Combinatorial Assignment.” *AAAI 2023*.
- Jakob Weissteiner*, Jakob Heiss*, Julien Siems*, Sven Seuken. “Monotone-Value Neural Networks: Exploiting Preference Monotonicity in Combinatorial Assignment.” *IJCAI-ECAI 2022*.
- Arber Zela*, Julien Siems*, Lucas Zimmer*, Jovita Lukasik, Margret Keuper, Frank Hutter. “Surrogate NAS Benchmarks: Going Beyond the Limited Search Spaces of Tabular NAS Benchmarks.” *ICLR 2022*.
- Julien Siems, Aaron Klein, Cédric Archambeau, Maren Mahsereci. “Dynamic Pruning of a Neural Network via Gradient Signal-to-Noise Ratio.” *8th ICML Workshop on Automated Machine Learning (AutoML), 2021*.
- Arber Zela*, Julien Siems*, Frank Hutter. “NAS-Bench-1Shot1: Benchmarking and Dissecting One-Shot Neural Architecture Search.” *ICLR 2020*.
- Roland Zimmermann*, Julien Siems*. “Faster Training of Mask R-CNN by Focusing on Instance Boundaries.” *Computer Vision and Image Understanding 188 (2019): 102795*.

PATENTS

- Arber Zela, Frank Hutter, Julien Siems, Lucas Zimmer. Method, device and computer program for predicting a suitable configuration of a machine learning system for a training data set. US Patent No. 16950570.
- Arber Zela, Frank Hutter, Julien Siems, Lucas Zimmer. Verfahren, Vorrichtung und Computerprogramm zum Vorhersagen einer geeigneten Konfiguration eines maschinellen Lernsystems für einen Trainingsdatensatz. German Patent No. DE: 10 2020 202 392.1.
- Julien Siems, Arber Zela, Frank Hutter. Verfahren, Vorrichtung und Computerprogramm zum Erstellen eines maschinellen Lernsystems. German Patent No. DE: 10 2019 214 500.0.